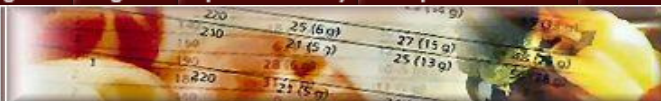


The 4-4-9 Method of Calculating Calorie Ratios



"Hell is the knowledge of opportunity lost." - Anonymous

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The "4-4-9" Method of Calculating Calorie Ratios - CyberSoft provides six fields of information that it calculated from existing nutrient data. These calculated fields include % *Calories from Protein*, % *Calories from Carbohydrates*, % *Calories from Fat*, % *Calories from Alcohol*, *Calories from Protein*, *Calories from Carbohydrates*, *Calories from Alcohol* and *Calories from Fat*.

NutriBase uses these values to derive the *PCF Ratio* that is displayed in all [recipes](#), nutrient [intakes](#), and [Meal Plans](#). These important numbers are derived in a variety of ways, depending on the type of nutrient data involved.

Because the 4-4-9 method is an official AOAC method of calculating calories, most laboratory analyses still report calories using this simple - *but crude* - method. This popular method of estimating calories assumes that one gram of protein contains 4 calories, that one gram of carbohydrate contains 4 calories, and that one gram of fat contains 9 calories. In beverages or foods containing alcohol, it is assumed that each gram of alcohol provides 7 calories. By knowing the total calories in the food and the total calories from the calorie sources contained in the food, it is a fairly straight-forward calculation to determine the "Calories from" and the "% Calories from" values.

For USDA data, which uses the [Atwater Conversion Factors](#) to determine "calories from" values for most food items (except for formulated foods), CyberSoft performed calculations based on the factors provided by the USDA. The [Atwater Conversion Factors](#) used by the USDA are known to be more accurate than the AOAC-approved 4-4-9 method. The Atwater Factors are conversion factors that take into consideration that different types of foods provide different amounts of food energy per unit of any given calorie source. These USDA calorie conversion factors are included in [NutriBase Clinical](#).

For USDA "formulated foods," which include multiple ingredients, CyberSoft used the AOAC-approved 4-4-9 method of calculating calorie ratios. CyberSoft's decision to not impose the [compensated 4-4-9 method](#) on USDA data produced some calorie ratio totals of less than or greater than 100%. (CyberSoft's policy regarding USDA data is to keep it "as-is." For this reason, all data from the original USDA source files have been retained unchanged.)

For food items listed in the [Vitamins](#) and Medical Nutritionals section of NutriBase (multiple [vitamins](#), [minerals](#), supplements, [enteral and parenteral products](#)), CyberSoft calculated calorie ratios by using the manufacturer's information regarding calorie percentages. (These values total 100%.)

The 4-4-9 method is - in theory - a workable method of calculating "% Calories from" values. However, when applied to published data from commercial [food manufacturers](#), this method often results in percentages not equal to 100%. (In fact, all too often, the calculated values are very far removed from 100%.) A somewhat more reliable method is the [compensated 4-4-9 method](#), which was pioneered by CyberSoft, Inc. in 1995.



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